Hyun-Jic Oh

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Research Summary

AI/ML researcher specializing in diffusion-based generative models for medical image analysis and data-efficient visual learning. My work develops image synthesis, translation, and augmentation methods to build robust medical AI systems under limited data, class imbalance, weak annotation, and modality gaps, with growing focus on multimodal AI and intelligent systems.

Experience

- **VIENCE Inc.** *Aug. 2026 – Present*
  AI Research Scientist Seoul, South Korea
  - Research and development of deep learning models for VIENCE PathoView, the company's pathology image analysis platform.
  - Currently improving a mitosis detection model for histopathology images.
- **Visual Computing Group, Harvard University** *April. 2024 – Mar. 2025*
  Visiting Scholar Cambridge, MA, USA
  - Developed marker-wise conditioned diffusion models for virtual multiplex staining, translating routine H&E histology into high-plex immunofluorescence images for spatially resolved tissue analysis. (AAAI 2026)
  - Collaborated with researchers at Harvard SEAS, HMS/MGH, and systems biology groups on computational pathology, multimodal biomedical imaging, and translational medical AI projects.
- **High Performance Visual Computing Lab, Korea University** *Aug. 2020 – Aug. 2026*
  Ph.D. in Computer Science Seoul, South Korea
  - Developed diffusion-based data synthesis and image-label co-generation pipelines for nuclei segmentation and classification under limited annotations, data scarcity, and class imbalance.
    (MICCAI 2023, MICCAI 2024, ECCV 2024)
  - Developed weakly supervised and prompt-efficient segmentation methods with contrastive regularization and foundation-model analysis.
    (ISBI 2022, CIBM 2025, MedAGI 2023)
  - Designed diffusion-guided 3D CT super-resolution using 2D projection priors and Signed 3D Gaussians.
    (MICCAI 2026 Early Accept, top ∼9%)
  - Developed diffusion augmentation method for data-scarce fine-grained classification using region-selective value attention and repulsive noise optimization.
  - Served as TA for Data Structures and Digital Image Processing; managed GPU servers; lab leader.
- **Medical Information Processing Laboratory, Korea University** *Mar. 2019 – Jun. 2019*
  Undergraduate Researcher Seoul, South Korea
  - Developed a virtual image reconstruction simulator for educational demonstrations in biomedical imaging.

Education

- **Korea University** *Mar. 2021 – Aug. 2026*
  Ph.D. in Computer Science Seoul, South Korea
  - GPA: 4.31/4.50 | Advisor: Won-Ki Jeong
  - Dissertation: *Controllable Diffusion-based Data Augmentation for Learning in Limited Data Regimes*
  - Ph.D. conferred Aug. 25, 2026
- **Korea University** *Mar. 2014 – Feb. 2021*
  B.S. in Biomedical Engineering Seoul, South Korea
  - GPA: 4.03/4.50
- **Chapman University** *Spring 2020*
  Exchange Student (Computer Science & Biostatistics) Orange, CA, USA
- **East China Normal University** *Summer 2016*
  KU–China Global Leadership Program Shanghai, China
  - HSK Level 4 (264/300, Dec. 2021)

Military Service

- **Republic of Korea Police** *Oct. 2016 – Jul. 2018*
  Seoul Metropolitan Police Agency Seoul, South Korea

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PREPRINT, W=WORKSHOP

[C.7] **Hyun-Jic Oh***, Gayeon Koh*, Won-Ki Jeong (2027). **Improving Diffusion-Based Augmentation for Data-Scarce Image Classification via Mask-Guided Reference Attention and Repulsive Noise Optimization**. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027)*, **to appear**.

[C.6] Jeonghyun Noh*, **Hyun-Jic Oh***, Won-Ki Jeong (2026). **Zero-shot CT Super-Resolution using Diffusion-based 2D Projection Priors and Signed 3D Gaussians**. In *Medical Image Computing and Computer Assisted Intervention (MICCAI 2026)*, ***Early Accept (top ∼9%)***.

[C.5] **Hyun-Jic Oh**, Junsik Kim, Zhiyi Shi, Yichen Wu, Yu-An Chen, Peter K. Sorger, Hanspeter Pfister, Won-Ki Jeong (2026). **Virtual Multiplex Staining for Histological Images using a Marker-wise Conditioned Diffusion Model**. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2026)*.

[C.4] **Hyun-Jic Oh**, Won-Ki Jeong (2024). **Controllable and Efficient Multi-Class Pathology Nuclei Data Augmentation using Text-Conditioned Diffusion Models**. In *Medical Image Computing and Computer Assisted Intervention (MICCAI 2024)*.

[C.3] Seonghui Min*, **Hyun-Jic Oh***, Won-Ki Jeong (2024). **Co-synthesis of Histopathology Nuclei Image-Label Pairs using a Context-Conditioned Joint Diffusion Model**. In *European Conference on Computer Vision (ECCV 2024)*.

[C.2] **Hyun-Jic Oh**, Won-Ki Jeong (2023). **DiffMix: Diffusion Model-based Data Synthesis for Nuclei Segmentation and Classification in Imbalanced Pathology Image Datasets**. In *Medical Image Computing and Computer Assisted Intervention (MICCAI 2023)*.

[C.1] **Hyun-Jic Oh**, Kanggeun Lee, Won-Ki Jeong (2022). **Scribble-Supervised Cell Segmentation Using Multiscale Contrastive Regularization**. In *IEEE International Symposium on Biomedical Imaging (ISBI 2022)*.

[J.2] **Hyun-Jic Oh***, Seonghui Min*, Won-Ki Jeong (2025). **S2L-CM: Scribble-supervised Nuclei Segmentation in Histopathology Images using Contrastive Regularization and Pixel-Level Multiple Instance Learning**. *Computers in Biology and Medicine*, 2025.

[J.1] Junyoung Choi, **Hyun-Jic Oh**, Hakjun Lee, Suyeon Kim, Seok-Kyu Kwon, Won-Ki Jeong (2023). **MitoVis: A Unified Visual Analytics System for End-to-End Neuronal Mitochondria Analysis**. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2023.

[P.1] **Hyun-Jic Oh**, Won-Ki Jeong (2026). **Virtual Multiplex Staining with Marker-Aware Decoding and Test-Time LoRA Scaling**. *Preprint*.

[W.3] **Hyun-Jic Oh** (2026). **Controllable Diffusion Models for Data Scarcity in Computational Pathology**. In *MICCAI 2026 Doctoral Symposium*, **selected for PhD Thesis Madness oral presentation**.

[W.2] Gayeon Koh*, **Hyun-Jic Oh***, Jeonghyun Noh, Won-Ki Jeong (2025). **Synthetic Data Augmentation using Pre-trained Diffusion Models for Long-tailed Food Image Classification**. In *2nd MetaFood Workshop at CVPR 2025*.

[W.1] Seungkyu Kim, **Hyun-Jic Oh**, Seonghui Min, Won-Ki Jeong (2023). **Evaluation and Improvement of Segment Anything Model for Interactive Histopathology Image Segmentation**. In *MICCAI 2023 1st International Workshop on Foundation Models for General Medical AI (MedAGI)*.

SKILLS

- **Programming Languages:** Python, MATLAB, C++
- **ML/AI Frameworks:** PyTorch, TensorFlow, MONAI, Hugging Face Diffusers
- **Imaging & Medical Data:** Histopathology, CT/MR imaging, multiplexed immunofluorescence, spatial omics, cellular/neuronal imaging
- **Tools & Platforms:** Linux, CUDA, Git, Docker, Slurm/HPC clusters, Weights&Biases, Hydra
- **Methods:** Diffusion models, generative modeling, data-centric augmentation, weakly supervised learning, contrastive learning, vision-language models, foundation models (SAM)

ACADEMIC SERVICES

- **Conference Reviewer**
  - AAAI (2026–2027), CVPR (2026), ECCV (2026), NeurIPS (2026), MICCAI (2023–2026), IEEE ISBI (2027), MICCAI MedAGI Workshop (2023–2024, 2026)
- **Journal Reviewer**
  - IEEE TMM, IEEE JBHI, Pattern Recognition (PR), Engineering Applications of AI (EAAI), Expert Systems with Applications (ESWA)
- **Program Committee**
  - Student Contributor, PAIP 2023 AI Challenge: TC prediction in pancreatic and colon cancer, IEEE ISBI 2023

Honors and Awards

- **PhD Thesis Madness (Oral Presentation)** *Sept. 2026*
  MICCAI 2026 Doctoral Symposium Strasbourg, France
  - Selected for the oral *PhD Thesis Madness* session, awarded to top-tier accepted participants of the Doctoral Symposium.

Additional Information

Languages: English (Fluent), Korean (Native), Chinese (HSK Level 4)
Research Interests: Diffusion-based generative models, multimodal medical AI, synthetic data generation and validation, data-efficient visual learning, weakly supervised segmentation, foundation models

References

1. **Prof. Won-Ki Jeong**
   Associate Professor, Department of Computer Science and Engineering
   Korea University, Seoul, South Korea
   Email: wkjeong@korea.ac.kr
   Relationship: Ph.D. advisor

2. **Prof. Hanspeter Pfister**
   An Wang Professor of Computer Science
   Harvard John A. Paulson School of Engineering and Applied Sciences
   Email: pfister@seas.harvard.edu
   Relationship: Host during visiting scholar period at Harvard VCG

3. **Dr. Junsik Kim**
   Applied Scientist, Amazon
   Email: mibastro@gmail.com
   Relationship: Postdoctoral host and direct research mentor during my visiting scholar period at Harvard VCG; co-author on the virtual multiplex staining project

4. **Dr. Yichen Wu**
   Research Fellow, Department of Systems Biology
   Harvard Medical School / Massachusetts General Hospital
   Email: yichen_wu@hms.harvard.edu
   Relationship: Collaborator on virtual multiplex staining project